

DBMS LAB # 5**Exercise # 1:****QUERY:**

```
SELECT Course.CourseID, Course.CourseName, Prog.ProgramName
FROM Course
JOIN Prog ON Course.ProgramID = Prog.ProgramID;
```

OUTPUT:

Results		Messages	
	CourseID	CourseName	ProgName
1	1	Calculus - 1	BSCS
2	2	Business Communications	BSCS
3	3	Introduction To Computing	BSCS
4	4	Calculus - 2	BSCS
5	5	Pak Studies	BSCS
6	6	Programming Fundamentals	BSCS
7	7	Discrete Maths	BSCS
8	8	Data Structures	BSCS
9	9	Digital Logic	BSCS
10	10	Visual Prog	BSCS
11	11	Computer Arch and Design	BSCS
12	12	Project Management	BSCS

Exercise # 2

QUERY:

```
SELECT Course.CourseID, Course.CourseName, Prog.ProgramName,  
Semester.SemesterName  
FROM Course  
JOIN Prog ON Course.ProgramID = Prog.ProgramID  
JOIN Semester ON Course.SemID = Semester.SemID;
```

OUTPUT:

110 %				
Results Messages				
	CourseID	CourseName	ProgName	SemName
1	1	Calculus - 1	BSCS	Spring2020
2	2	Business Communications	BSCS	Spring2020
3	3	Introduction To Computing	BSCS	Spring2020
4	4	Calculus - 2	BSCS	Summer2020
5	5	Pak Studies	BSCS	Summer2020
6	6	Programming Fundamentals	BSCS	Summer2020
7	7	Discrete Maths	BSCS	Fall2020
8	8	Data Structures	BSCS	Fall2020
9	9	Digital Logic	BSCS	Spring2021
10	10	Visual Prog	BSCS	Spring2021
11	11	Computer Arch and Design	BSCS	Spring2021
12	12	Project Management	BSCS	Spring2021

Exercise # 3

QUERY:

```
SELECT Student.StdID, Student.StdName
FROM Student
JOIN Prog ON Student.ProgramID = Prog.ProgramID
WHERE Prog.ProgramName <> 'BSCS';
```

OUTPUT:

Results Messages	
StdID	StdName

Exercise # 4

QUERY:

```
SELECT Teacher.TeacherName, Course.CourseName
FROM Teacher
JOIN StdCourseTeacher ON Teacher.TeacherID =
StdCourseTeacher.TeacherID
JOIN Course ON StdCourseTeacher.CourseID = Course.CourseID
WHERE Teacher.TeacherType = 'Visiting';
```

OUTPUT:

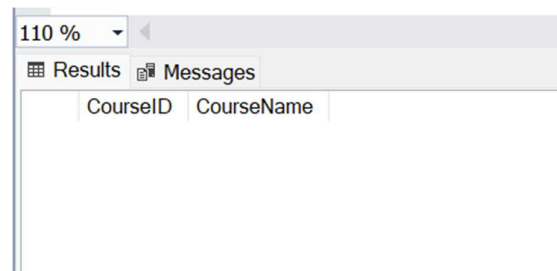
Results Messages		
	TeacherName	CourseName
1	Dr. Iman	Business Communications
2	Dr. Iman	Pak Studies
3	Dr. Khalid	Calculus - 1
4	Dr. Khalid	Calculus - 2
5	Dr. Khalid	Discrete Maths

Exercise # 5

QUERY:

```
SELECT Course.CourseID, Course.CourseName
FROM Course
JOIN StdCourseTeacher ON Course.CourseID =
StdCourseTeacher.CourseID
JOIN Teacher ON StdCourseTeacher.TeacherID = Teacher.TeacherID
JOIN Semester ON StdCourseTeacher.SemID = Semester.SemID
WHERE Teacher.TeacherName = 'Dr. Iman' AND
Semester.SemesterName = 'Fall2023';
```

OUTPUT:



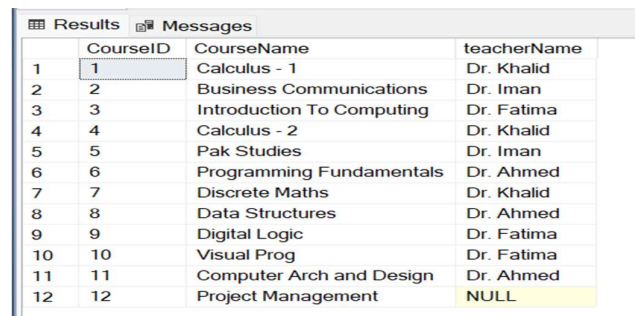
CourseID	CourseName
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Exercise # 6

QUERY:

```
SELECT Course.CourseID, Course.CourseName, Teacher.TeacherName
FROM Course
LEFT JOIN StdCourseTeacher ON Course.CourseID =
StdCourseTeacher.CourseID
LEFT JOIN Teacher ON StdCourseTeacher.TeacherID =
Teacher.TeacherID;
```

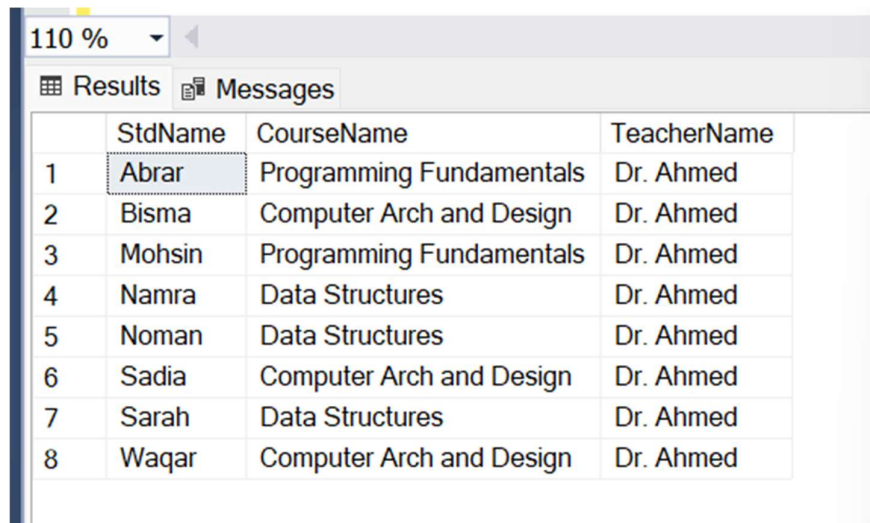
OUTPUT:



CourseID	CourseName	teacherName
1	Calculus - 1	Dr. Khalid
2	Business Communications	Dr. Iman
3	Introduction To Computing	Dr. Fatima
4	Calculus - 2	Dr. Khalid
5	Pak Studies	Dr. Iman
6	Programming Fundamentals	Dr. Ahmed
7	Discrete Maths	Dr. Khalid
8	Data Structures	Dr. Ahmed
9	Digital Logic	Dr. Fatima
10	Visual Prog	Dr. Fatima
11	Computer Arch and Design	Dr. Ahmed
12	Project Management	NULL

Exercise # 7**QUERY:**

```
SELECT Student.StdName, Course.CourseName
FROM Student
JOIN StdCourseTeacher ON Student.StdID =
StdCourseTeacher.StdID
JOIN Course ON StdCourseTeacher.CourseID = Course.CourseID
JOIN Teacher ON StdCourseTeacher.TeacherID = Teacher.TeacherID
WHERE Teacher.TeacherName = 'Dr. Ahmed';
```

OUTPUT:A screenshot of the SQL Server Enterprise Manager interface. At the top, there is a zoom level dropdown set to '110 %' and a back arrow. Below this are two tabs: 'Results' (active) and 'Messages'. The 'Results' tab displays a table with four columns: 'StdName', 'CourseName', and 'TeacherName'. There are 8 rows of data. The first row is highlighted with a blue selection bar. The data is as follows:

	StdName	CourseName	TeacherName
1	Abrar	Programming Fundamentals	Dr. Ahmed
2	Bisma	Computer Arch and Design	Dr. Ahmed
3	Mohsin	Programming Fundamentals	Dr. Ahmed
4	Namra	Data Structures	Dr. Ahmed
5	Noman	Data Structures	Dr. Ahmed
6	Sadia	Computer Arch and Design	Dr. Ahmed
7	Sarah	Data Structures	Dr. Ahmed
8	Waqar	Computer Arch and Design	Dr. Ahmed

Exercise # 8

QUERY:

```

SELECT Student.StdID, Student.StdName, Course.CourseID,
Course.CourseName,
        Teacher.TeacherID, Teacher.TeacherName, Semester.SemID,
Semester.SemesterName
FROM Student
JOIN StdCourseTeacher ON Student.StdID =
StdCourseTeacher.StdID
JOIN Course ON StdCourseTeacher.CourseID = Course.CourseID
JOIN Teacher ON StdCourseTeacher.TeacherID = Teacher.TeacherID
JOIN Semester ON StdCourseTeacher.SemID = Semester.SemID;

```

OUTPUT:

	StdID	StdName	TeacherID	TeacherName	CourseID	CourseName	SemID	SemName
1	1	Bisma	7	Dr. Fatima	9	Digital Logic	6	Fall2021
2	1	Bisma	7	Dr. Fatima	10	Visual Prog	6	Fall2021
3	1	Bisma	2	Dr. Ahmed	11	Computer Arch and Design	6	Fall2021
4	2	Sadia	7	Dr. Fatima	9	Digital Logic	6	Fall2021
5	2	Sadia	7	Dr. Fatima	10	Visual Prog	6	Fall2021
6	2	Sadia	2	Dr. Ahmed	11	Computer Arch and Design	6	Fall2021
7	3	Sarah	3	Dr. Khalid	7	Discrete Maths	4	Spring2021
8	3	Sarah	2	Dr. Ahmed	8	Data Structures	4	Spring2021
9	4	Namra	3	Dr. Khalid	7	Discrete Maths	4	Spring2021
10	4	Namra	2	Dr. Ahmed	8	Data Structures	4	Spring2021
11	5	Waqar	7	Dr. Fatima	9	Digital Logic	6	Fall2021
12	5	Waqar	7	Dr. Fatima	10	Visual Prog	6	Fall2021
13	5	Waqar	2	Dr. Ahmed	11	Computer Arch and Design	6	Fall2021
14	6	Noman	3	Dr. Khalid	7	Discrete Maths	4	Spring2021
15	6	Noman	2	Dr. Ahmed	8	Data Structures	4	Spring2021
16	7	Abrar	3	Dr. Khalid	4	Calculus - 2	3	Fall2020
17	7	Abrar	4	Dr. Iman	5	Pak Studies	3	Fall2020
18	7	Abrar	2	Dr. Ahmed	6	Programming Fundament...	3	Fall2020
19	8	Mohsin	3	Dr. Khalid	4	Calculus - 2	3	Fall2020
20	8	Mohsin	4	Dr. Iman	5	Pak Studies	3	Fall2020
21	8	Mohsin	2	Dr. Ahmed	6	Programming Fundament...	3	Fall2020
22	9	Sikandar	3	Dr. Khalid	1	Calculus - 1	6	Fall2021
23	9	Sikandar	4	Dr. Iman	2	Business Communications	6	Fall2021
24	9	Sikandar	7	Dr. Fatima	3	Introduction To Computing	6	Fall2021
25	10	Javed	3	Dr. Khalid	1	Calculus - 1	6	Fall2021
26	10	Javed	4	Dr. Iman	2	Business Communications	6	Fall2021

Exercise # 9**QUERY:**

```
SELECT DISTINCT Teacher.TeacherID, Teacher.TeacherName
FROM Teacher
JOIN StdCourseTeacher ON Teacher.TeacherID =
StdCourseTeacher.TeacherID
JOIN Semester ON StdCourseTeacher.SemID = Semester.SemID
WHERE Semester.SemesterName = 'Fall2023';
```

OUTPUT:

Results Messages	
TeacherID	TeacherName

Exercise # 10**QUERY:**

```
SELECT DISTINCT Teacher.TeacherID, Teacher.TeacherName
FROM Teacher
JOIN StdCourseTeacher ON Teacher.TeacherID =
StdCourseTeacher.TeacherID;
```

OUTPUT:

Results Messages		
	TeacherID	TeacherName
1	2	Dr. Ahmed
2	3	Dr. Khalid
3	4	Dr. Iman
4	7	Dr. Fatima

Exercise # 11**QUERY:**

```

SELECT Teacher.TeacherID, Teacher.TeacherName,
Course.CourseName
FROM Teacher
LEFT JOIN StdCourseTeacher ON Teacher.TeacherID =
StdCourseTeacher.TeacherID
LEFT JOIN Course ON StdCourseTeacher.CourseID =
Course.CourseID;

```

OUTPUT:

	TeacherID	TeacherName	CourseName
1	2	Dr. Ahmed	Computer Arch and Design
2	2	Dr. Ahmed	Data Structures
3	2	Dr. Ahmed	Programming Fundamentals
4	3	Dr. Khalid	Calculus - 1
5	3	Dr. Khalid	Calculus - 2
6	3	Dr. Khalid	Discrete Maths
7	4	Dr. Iman	Business Communications
8	4	Dr. Iman	Pak Studies
9	5	Dr. Safdar	NULL
10	6	Dr. Sohaib	NULL
11	7	Dr. Fatima	Digital Logic
12	7	Dr. Fatima	Introduction To Computing
13	7	Dr. Fatima	Visual Prog

Exercise # 12**QUERY:**

```

SELECT Teacher.TeacherID, Teacher.TeacherName
FROM Teacher
LEFT JOIN StdCourseTeacher ON Teacher.TeacherID =
StdCourseTeacher.TeacherID
WHERE StdCourseTeacher.TeacherID IS NULL;

```

OUTPUT:

	TeacherID	TeacherName	CourseID	SemID
1	5	Dr. Safdar	NULL	NULL
2	6	Dr. Sohaib	NULL	NULL

Exercise # 13**QUERY:**

```
SELECT Semester.SemID, Semester.SemesterName,  
COUNT(Student.StdID) AS EnrolledStudents  
FROM Semester  
LEFT JOIN StdCourseTeacher ON Semester.SemID =  
StdCourseTeacher.SemID  
LEFT JOIN Student ON StdCourseTeacher.StdID = Student.StdID  
WHERE Student.StdID IS NULL  
GROUP BY Semester.SemID, Semester.SemesterName;
```

OUTPUT:

Results		Messages	
	SemID	SemName	EnrolledStudents
1	1	Spring2020	0
2	2	Summer2020	0
3	5	Summer2021	0